Opening

A solar site can look promising until the documents disagree.

Project teams reconcile land titles, protection diagrams, land conditions and financial assumptions before making expensive engineering decisions. Heliopolis-AI combines AI document extraction with deterministic checks to flag issues requiring human review earlier.

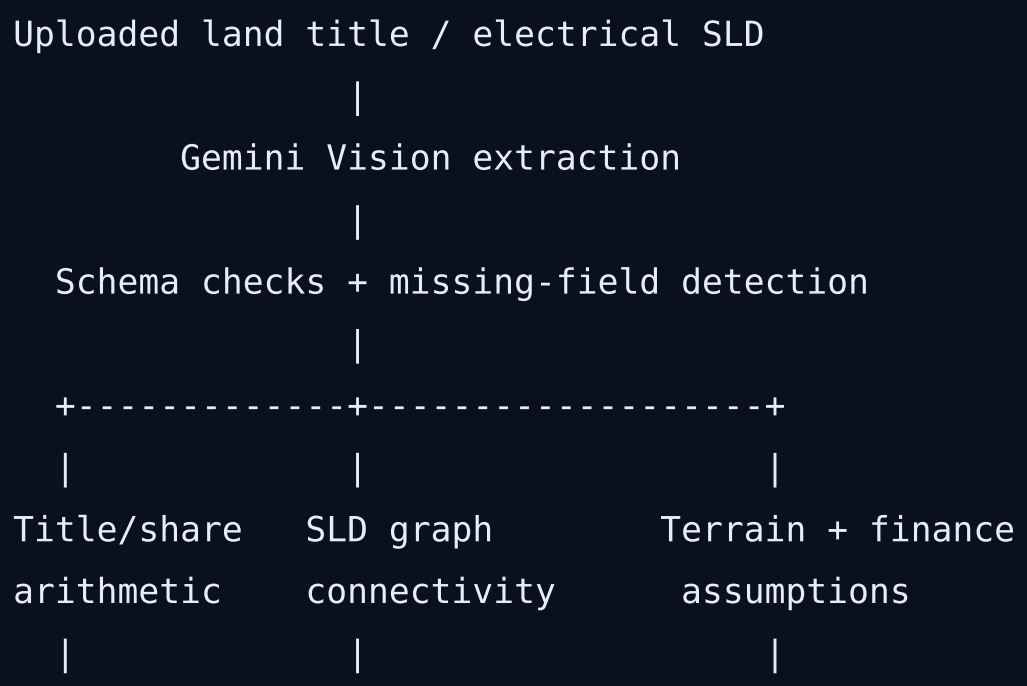
One-line product: "Preliminary solar-site screening with traceable calculations and explicit uncertainty."

The user's problem

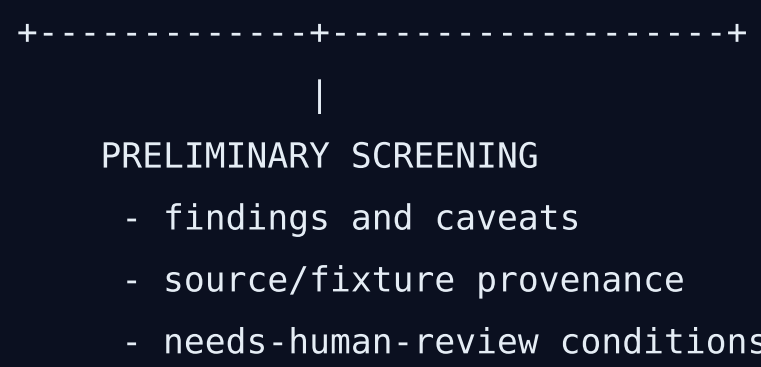
Solar project-development analysts receive different file types, inconsistent fields and site assumptions. A wrong land area, overlooked encumbrance, omitted protection relay or unsuitable financing scheme can change an early screening decision.

Pilot hypothesis — not a validated fact: structured extraction plus auditable checks can reduce manual reconciliation and help identify problems earlier. Time saved and false-clearance rates must be measured on real, permitted project documents.

System architecture (part 1/2)



System architecture (part 2/2)



AI's exact role: multimodal perception. Python's role: unit conversions, ownership share sums, topology traversal, yield/cash-flow arithmetic. No trained Heliopolis foundation model is claimed.

Product demonstration: three outcomes

1. Clearly labeled synthetic scenario: show the input, selected finance scenario, numerical outputs, SLD topology and review notes.
2. Faulty SLD: a present-but-disconnected protection relay must yield NEEDS_REVIEW rather than a confident grid-compliance claim.
3. Failed extraction: missing Gemini credentials, unreadable legal restrictions or malformed SLD geometry must stop the uploaded-document assessment, not silently produce a compliant demo fixture.

The demonstration is not a live Malaysian land-office or TNB integration.

Engineering scope and limits

- Land: preliminary field and owner-share consistency; no claim that 51% share automatically authorizes lease execution.
- SLD: check modeled PV -> inverter -> anti-islanding relay -> transformer -> grid connectivity. A graph pass is not a TNB approval, relay-settings study or electrical certification.
- Terrain: generated elevation profiles are synthetic, not surveyed or lidar-derived project terrain.
- Finance: CAPEX, loss factors, PPA/offset rates, WACC and yield assumptions are scenario inputs, not bankable quotes.
- BESS/grid: simple AC-MW / upstream-MVA ratio is a sensitivity screen, not a load-flow study or mandatory 15% battery trigger.

Regulations matter: do not mix schemes

LSS PPA: standalone utility-scale economic sensitivity, subject to applicable procurement and connection requirements.

NEM NOVA / illustrative C&I offset: separate earlier scheme; current eligibility and export credit must be verified independently. Rate assumptions are not customer-specific electricity-bill calculations.

Solar ATAP (2026): not implemented as a complete eligibility/financial model. The published non-domestic rule caps inverter output at 1 MW and 100% of consumer maximum demand; export-credit arrangement is 10 years, followed by self-consumption without export offsets. Do not present the example ~5.9 MWac system as Solar ATAP eligible. Actual offsets require load/export profiles and verified tariff components.

Authorities:

Primary references: official SEDA Malaysia ATAP and NEM programme guidance.

Evidence, not inflated benchmarks

What exists: an executable Python/Streamlit prototype, unit/integration tests and reproducible synthetic fixtures. The repo's CI runs tests and performs Python compilation.

What is not yet demonstrated: field OCR accuracy on held-out legal records, SLD error rates on real drawings, time savings against a manual process, utility validation, Solarvest partnership, or revenue.

Next empirical pilot: 3–5 consented real projects, blinded expert labels, comparison with direct Gemini extraction and ordinary spreadsheet checks, measurement of correction time, false-clearance rate, abstention, accuracy and latency. Do not score "needs review" as a verified negative.

Customer and commercial validation

Potential users: solar EPC project developers, IPPs and due-diligence analysts.

Starting integration: project-development analyst uploads title/SLD and enters proposed technical assumptions, receives a preliminary issue report before engaging a surveyor, conveyancing lawyer and power-systems engineer.

Proposed business model (unvalidated): enterprise screening/workflow subscription or integration. Test willingness to pay and time saved during a pilot. Do not claim signed customer contracts.

Final ask

We seek an industry pilot partner to validate the accuracy and decision value of preliminary screening on 3–5 real Malaysian solar projects. The pilot will define measurable go/no-go criteria before adding registry integration or automatic application drafting.

Closing: "Heliopolis-AI does not replace the engineers. It makes early assumptions visible, repeatable and reviewable."